

Auishik Pyne

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in generative AI, autonomous agent systems, and production ML pipelines. Experienced in LLM applications with RAG architectures, self-hosted agent frameworks with persistent memory and tool-use, and scalable speech/audio ML deployment. Strong software engineering foundation (Python, cloud infrastructure, ML lifecycle). MS CS candidate at Georgia Tech.

SKILLS

AI & ML: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Predictive Modeling, Prompt Engineering, RAG, LangChain/LlamaIndex, FAISS, Vector Databases, ChromaDB, Fine-Tuning, MLflow, DVC, TorchServe

Agent Systems: Autonomous Agents, Tool-Use / Function Calling, Ollama, n8n, LLM Adapter Layers, Memory Systems, Multi-Agent Orchestration

Languages: Python, SQL, C, C++, Java, JavaScript, HTML/CSS, LaTeX

Cloud & DevOps: AWS, GCP, Lambda Labs, Docker, FastAPI, Flask, WebSockets, Redis, MongoDB, ONNX, Git, Linux

Data & Analytics: Streamlit, Power BI, Tableau, Matplotlib, Seaborn, Statistical Analysis, Data Visualization, A/B Testing

EXPERIENCE

Data Quality Analyst

Dec 2025 - Present

Meta (via Experis) | Redmond, WA

- Evaluated multimodal and voice Meta AI LLM responses, maintaining >95% quality accuracy across thousands of data points for model alignment and benchmark regression tracking.
- Assessed complex reasoning tasks for Meta AI Glasses, identifying model quality gaps to improve on-device AI performance.
- Audited Bengali-English conversational translations for Meta AI training pipelines, sustaining >96% contextual accuracy and linguistic fidelity.
- Led peer review sessions supporting DQA certification, strengthening team-wide annotation consistency and performance metrics.

Machine Learning Engineer

Apr 2022 - Nov 2024

REVE Systems | Dhaka, Bangladesh

- Architected a tri-layered biometric authentication system (speaker, facial, fingerprint) achieving near-perfect accuracy with ROC/AUC analysis for precision-recall optimization.
- Trained and deployed speech-based gender and age prediction models achieving 93% and 74% accuracy, with data drift monitoring and model lifecycle management.
- Built a multi-speaker TTS system with custom data pipelines, boosting inference speed 3x via asynchronous batch processing and caching.
- Integrated speaker diarization pipeline achieving >80% accuracy in multi-party conversations; conducted unsupervised clustering (K-means, t-SNE) for data augmentation.
- Improved VAD accuracy 2x via noise reduction; accelerated audio processing 5x with Python multiprocessing. Integrated Azure and GCP STT/TTS APIs.
- Developed a Bengali text normalizer and FastSpeech 2 TTS model deployed via Docker and Flask.

QA Specialist

Mar 2025 - Nov 2025

PROSRVC LLC | Bothell, WA

- Conducted quality control and functional testing on 100+ devices daily, ensuring compliance with technical standards and performance benchmarks.
- Analyzed testing errors to identify systemic patterns, improving diagnostic accuracy to >96%.

Software Development Engineer Intern

May 2019 - Jun 2019

SELISE Rockin' Software | Dhaka, Bangladesh

- Developed a full-stack quiz platform with timed assessments, performance tracking, and an administrative analytics dashboard.

PROJECTS

Self-Hosted AI Agent - Second Brain

FastAPI, ChromaDB, n8n, Ollama

Modular autonomous agent managing tasks, calendar, email, and knowledge. Features a hot-swappable LLM adapter layer (DeepSeek/Ollama <-> Claude/GPT-4) and ChromaDB vector memory for persistent semantic recall across sessions.

AI Data Analytics Agent

Streamlit, LLM APIs

Conversational agent enabling non-technical users to query datasets in natural language, receiving visualizations, insights, and actionable summaries without writing code.

The Medical Bill Crusher

DeepSeek V3, Tesseract OCR

AI-powered medical bill auditing pipeline that automates discrepancy detection between bills and EOBs, generating regulatory appeal scripts to catch billing errors.

Interactive PDF Chat Application

Mistral-7B, FAISS

Source-aware RAG pipeline with advanced chunking and semantic search techniques, minimizing hallucinations and improving context retrieval accuracy for document Q&A.

EDUCATION

Master of Science in Computer Science

Expected Aug 2027

Georgia Institute of Technology | Specialization: Machine Learning

Bachelor of Science in Electrical & Computer Engineering

Sep 2021

Rajshahi University of Engineering & Technology (RUET)

PUBLICATION

- A. Pyne, S. Anower. "Performance Comparison of Multiple Supervised ML Algorithms for COVID-19 Mortality Prediction" — ICAECT 2022, IEEE. Achieved 95% accuracy and 89% F1-score using ensemble ML methods on the Johns Hopkins University dataset.